

MATH 1342 – Elementary Statistics – Week 10: A Closer Look at Testing, Making Connections, and Normal Distributions

Oct 26 – 28 | 4.4(a) A Closer Look at Testing • 4.5 Making Connections • 5.1 Hypothesis Testing Using Normal Distributions | Fall 2026 • Section 019 • MW 7:30–8:50 PM
• RRC Room 8210

Course-level Objectives	Module-level Objectives	Learning Materials	#	Activities	#	Assessments	#
3. Find and use empirical probabilities in bootstrap distributions to find confidence intervals and in randomization distributions to test hypotheses. 4. Find and use theoretical probabilities from normal, t, chi-squared and F distributions to form confidence intervals and test hypotheses. Apply the 95% rule to normal and to approximately normal distributions. 8. Use the sampling distributions of sample proportions and sample means to answer appropriate questions. 9. Estimate single means, difference of two means, single proportions and difference of two proportions using confidence intervals. Interpret the results. 10. Demonstrate skills in hypothesis testing for means and proportions, for single populations and comparison of two populations.	1. Describe Type I and Type II errors in the context of a study, relate α to the probability of a Type I error, and explain the trade-off between the two errors and the effect of sample size. (CO10) 2. Explain how multiple testing and publication bias produce false positives, and distinguish statistical significance from practical significance. (CO10) 3. Describe how to generate randomization samples for a given study design (coin flips, re-randomizing groups, shifting data then bootstrapping), and compare bootstrap and randomization distributions by their center and use. (CO3) 4. Use a 95% confidence interval to reach a two-tailed test conclusion at $\alpha = 0.05$, and decide whether a confidence interval or a hypothesis test fits a research question. (CO9, CO10) 5. Explain when the Central Limit Theorem justifies replacing a simulated randomization distribution with a normal distribution $N(\text{null value}, SE)$. (CO4, CO8) 6. Compute a standardized z statistic and find the p-value as a right-tail, left-tail, or two-tail area of the standard normal distribution. (CO4, CO10)	e-Text: 4.4 A Closer Look at Testing	MO1, MO2	Participation: Errors in Testing – battery-life example (a)–(d): describe each error in context; courtroom analogy matching; vitamin E multiple-testing questions	MO1, MO2	Online Homework: 4.4 A Closer Look at Testing (a) (due Sun Nov 1, 11:59 PM)	MO1, MO2
		Interactive Lecture: 4.4 A Closer Look at Testing	MO1, MO2			Online Homework: 4.4 A Closer Look at Testing (b) (due Sun Nov 1, 11:59 PM)	MO1, MO2
		Slides: 4.4 A Closer Look at Testing (includes Khan Academy Type I/II video)	MO1, MO2			Online Homework: 4.5 Making Connections (due Sun Nov 1, 11:59 PM)	MO3, MO4
		e-Text: 4.5 Making Connections	MO3, MO4	Participation: Making Connections – body temperature: test 98.6 using the CI (98.05, 98.47); sleep study (28/50): how to generate randomization samples	MO3, MO4	Online Homework: 4.5 Making Connections (due Sun Nov 1, 11:59 PM)	MO3, MO4
		Interactive Lecture: 4.5 Making Connections	MO3, MO4			Online Homework: 5.1 Hypothesis Tests Using Normal Distributions (due Sun Nov 1, 11:59 PM)	MO5, MO6
		Slides: 4.5 Making Connections	MO3, MO4			Test 2 – Fri Oct 30 – Sun Nov 1 (Blackboard, proctored, 2 hours). Covers Weeks 6–9.	None
		e-Text: 5.1 Hypothesis Tests Using Normal Distributions	MO5, MO6	Participation: Normal Distributions and z – malaria/mosquito example: compute z, shade the tail in StatKey, find the p-value	MO5, MO6	Take-Home Quiz 6 (due Mon Nov 2, 11:59 PM)	All
		Interactive Lecture: 5.1 Hypothesis Tests Using Normal Distributions	MO5, MO6				
		Slides: 5.1 Hypothesis Tests Using Normal Distributions	MO5, MO6				
		StatKey – Theoretical Distributions (Normal)	MO6				